

Diabetes Prediction Project

Submission Details

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Project Overview

This project develops a machine-learning classifier for predicting diabetes from patient health and lifestyle information.

The final model is stored with its fitted preprocessing transformer and is used by the Streamlit prediction interface.

Dataset

The dataset is diabetes_prediction_dataset.csv. The target column is diabetes.

Input variables include gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level, and blood glucose level.

The dataset was checked for missing values and duplicate rows. Duplicate records were removed before modeling.

Dataset Workflow

Preparation

The data is loaded with pandas, duplicate rows are removed, and the index is reset.

Two derived variables are created: `cardio_risk`, based on hypertension and heart disease, and `age_bmi`, based on age multiplied by BMI.

Preprocessing

The data is split into training and testing subsets using an 80/20 split with `random_state=42`.

Numerical features are standardized with `StandardScaler`. Categorical features are converted using `OneHotEncoder` with unknown categories ignored.

A `ColumnTransformer` keeps preprocessing consistent during training and prediction.

Training Workflow

Five models are trained and evaluated. `GridSearchCV` uses five-fold cross-validation and F1 scoring for hyperparameter tuning.

The selected model and the fitted preprocessor are saved together in `best_diabetes_model.pkl`.

Model Selection and Results

Models Compared

Decision Tree, Random Forest, XGBoost, Support Vector Machine, and Gradient Boosting were compared.

F1 score was used as the primary selection metric because it balances precision and recall for the positive diabetes class.

Selected Model

The saved model is GradientBoostingClassifier. It was selected as the best model according to the F1-score comparison.

Evaluation Results

The saved model was evaluated on the held-out test set:

Accuracy: 0.9710

Precision: 0.9866

Recall: 0.6851

F1 score: 0.8086

ROC-AUC: 0.9790

Limitations and Future Improvements

Limitations

The dataset may not represent every population or clinical setting.

A single train-test split can produce results that vary with the selected random split.

Duplicate removal does not correct measurement errors, missing context, or biased labels.

The prediction is a machine-learning estimate and must not replace professional medical diagnosis.

The current interface does not provide calibrated risk probabilities or explanations for individual predictions.

Future Improvements

Use repeated stratified cross-validation and report confidence intervals.

Investigate class weighting or resampling if class imbalance affects recall.

Calibrate probability estimates and add explainability with feature importance or SHAP.

Validate the model on an independent external dataset.

Add stronger input validation, model versioning, monitoring, and retraining procedures.